HDB Rental Rate Prediction

CS5228 Progress Report · Working Draft

26 September 2026

Problem, Goals and Research Questions

Task: predict monthly rent for an HDB rental approval record from property attributes and information available at the prediction cutoff.

Goal: support rental comparisons and evaluate generalization to future months. The target is `MONTHLY_RENT`, in **SGD/month**.

1. How does rent differ across property characteristics?
2. Can models generalize across future months as rent levels change?
3. Do historically available auxiliary features improve prediction?

Metric: original-scale RMSE in SGD/month; lower is better.

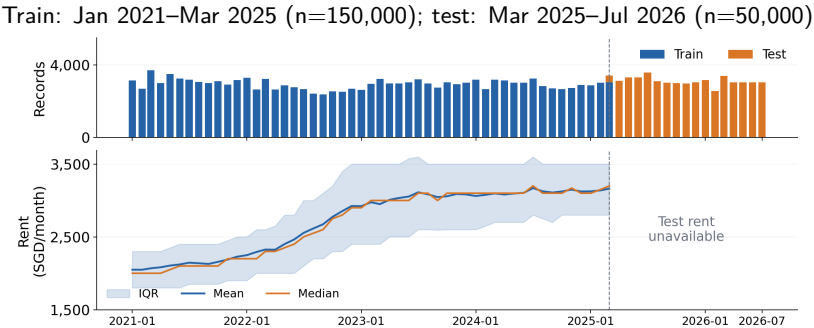
$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}$$

Data and Prediction Setting

Split	Records	Approval months	Columns
Train	150,000	Jan 2021–Mar 2025	10 inputs + rent
Test	50,000	Mar 2025–Jul 2026	10 inputs, no rent

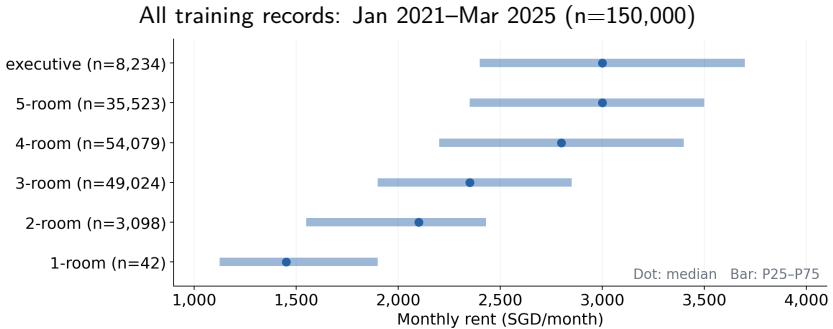
- **Inputs:** month, address, flat type/model, area and lease year, plus two constants (`FURNISHED=yes`, `FEE=0`). Core fields contain no parsed missing values or blank strings.
- **Prediction setting:** training ends in March 2025; test extends up to **16 months** later. The same month also contains **380 test rows**, with unknown within-month order.
- **Audit scope:** all core records and six auxiliary tables. No unit identifier is supplied, and no test rent is available.

EDA: Rent Levels and Changes over Time



- Training median: **2,700 SGD/month**; P25–P75: **2,100–3,200**; P1–P99: **1,300–4,600**.
- **99.24%** of test rows follow the cutoff, at horizons up to **16 months**. Observed rent changes do not establish the future trend.

EDA: Rent Differences across Properties



- Town rent distributions overlap and sample sizes differ. Town and flat-type comparisons mix property characteristics and time.
- Rent correlates with **area (+0.329)** and **lease-based age (−0.178)**. These marginal associations do not establish causal effects.

EDA: Data Quality and Auxiliary Readiness

Audit	Evidence	Proposed handling
Categories	Training categories: Streets: **1,146 to 576** Flat types: **11 to 6**	Normalize spelling; preserve block suffixes.
Repeated inputs	Same inputs, different rents **17,900 groups** **39,540 training rows**	Retain records; duplication cannot be confirmed without unit or contract IDs.
Normalized HDB join	**100%** of train and test rows matched **Zero** ambiguous keys	Validate many-to-one joins; check missing attributes.

Auxiliary timing: all six sources were audited. Facility snapshots lack effective dates; future observed COE or stock values cannot become prediction inputs.

Feature Design and Data Preparation

Outline — pending completion

Owners: A / B / C

- Translate the EDA into an agreed cleaning and feature specification, including building and calendar inputs.
- Specify auxiliary admission rules, unknown-value handling and preprocessing fitted within each training fold.
- Add one data-flow diagram from source records to legal model inputs.

Candidate Methods and Priorities

Outline — pending completion

Owners: A / B / C

- Summarize the common baselines and modern candidates in one compact comparison.
- Connect each candidate to an EDA-motivated hypothesis and explain the proposed attempt order.
- State resource assumptions and the evidence needed to retain or reject a candidate.

Temporal Validation and Evaluation

Outline — pending completion

Owners: B / C

- Add the agreed temporal split diagram, with training cutoffs and forecast horizons.
- Define full-window RMSE and horizon, tail and property-group diagnostics.
- Document preprocessing boundaries, tuning rules and overlap or holdout limitations.

Experiment Design and Ablations

Outline — pending completion

Owners: B / C

- Add one comparison matrix covering common baselines, feature additions and temporal variants.
- Identify the hypothesis tested by each experiment and the factors held constant.
- Specify experiment IDs, shared budgets and the results to record.

Progress and Preliminary Results

Outline — pending completion

Owners: B / C / D

No model results are available yet.

- Summarize actual execution status and link completed work to reproducible records.
- Once experiments run, report IDs, validation RMSE, runtime/resource use and the matching baseline.
- Explain findings and failed attempts using measured evidence.

Challenges and Open Decisions

Outline — pending completion

Owners: A / B / C / D

- Select the most consequential obstacles: temporal extrapolation, auxiliary history, data boundaries and resources.
- State each obstacle's impact, proposed resolution and responsible owner.
- Separate unresolved data questions from observed experiment failures.

Next Steps: Model Development

Outline — pending completion

Owners: A / B / C / D

- Plan work after the progress report: common baselines, prioritized candidate tests and feature ablations.
- Define the error analyses and evidence needed before selecting a model.
- Assign an owner and concrete deliverable to each development milestone.

Next Steps: Final Evaluation and Delivery

Outline — pending completion

Owners: A / B / C / D

- Plan final model confirmation and any justified ensemble comparison under the agreed protocol.
- Schedule authorized Kaggle checks, reproducibility packaging and final-report preparation.
- Define final review and submission responsibilities once the project deadline is confirmed.